

Executive Summary – Titanic Survival Prediction

Project Objective

The objective of this project was to predict passenger survival on the Titanic using demographic and travel-related information while emphasizing feature engineering and model explainability.

Dataset

The Titanic dataset contains passenger information including age, sex, ticket class, fare, cabin details, and family relationships.

Feature Engineering

Several new features were created:

- Title Extraction from passenger names
- Family Size calculation
- Cabin Presence indicator

These engineered features improved model performance and interpretability.

Missing Value Handling

Missing values were handled using:

- Median imputation for Age and Fare
- Mode imputation for Embarked
- Cabin information transformed into a binary feature

Models Evaluated

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

Evaluation Metrics

Models were evaluated using:

- Accuracy

- Precision
- Recall
- F1 Score
- Confusion Matrix

Explainability

Feature importance analysis revealed that Sex, Passenger Class, Fare, Age, and engineered features such as Title and Family Size were among the most influential predictors.

Results

The selected model achieved strong predictive performance and demonstrated the value of feature engineering in improving classification accuracy.

Conclusion

This project successfully combined data preprocessing, feature engineering, classification modeling, and explainability techniques to predict Titanic passenger survival. The final model was saved and can be used for future inference tasks.